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PLANISUS



Ecosystem Simulation



**Every morning in Planisuss, a Erbast wakes up.
It knows it must outrun the fastest Carviz or it will be killed.**

**Every morning in Planisuss, a Carviz wakes up.
It knows it must run faster than the slowest Erbast, or it will starve.**

**It doesn't matter whether you're the Carviz or a Erbast –
when the sun comes up, you'd better be running.
Unless you are a Vegetob.**

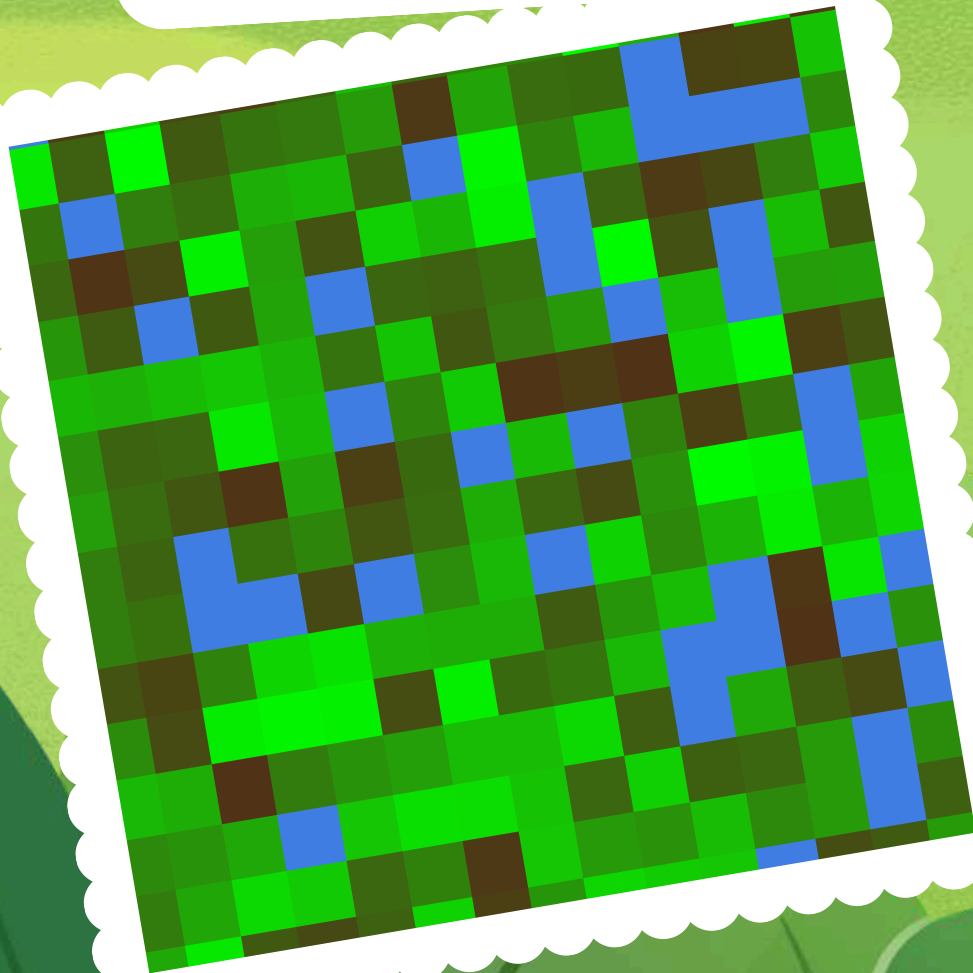
Planisuss proverb

What is Planisuss?

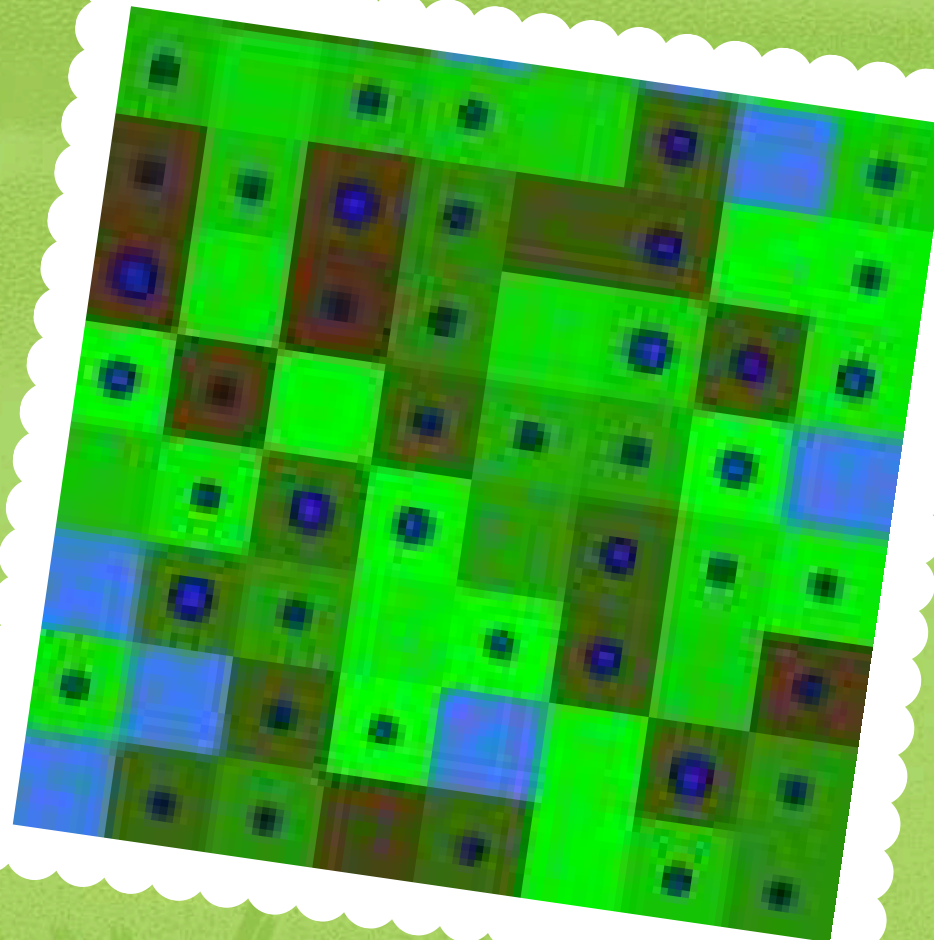


**A small artificial world
populated by Vegetob,
Erbast and Carviz.**

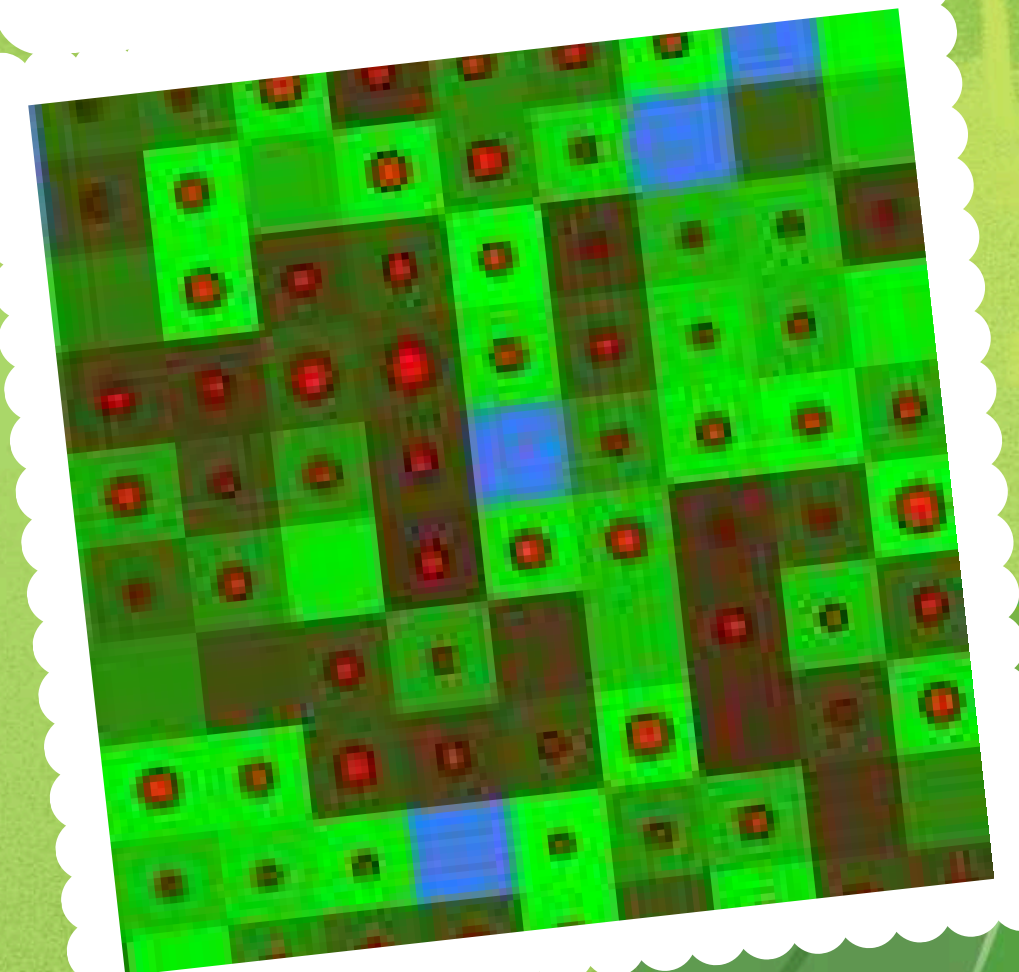
VEGETOB:
Food for ERBAST



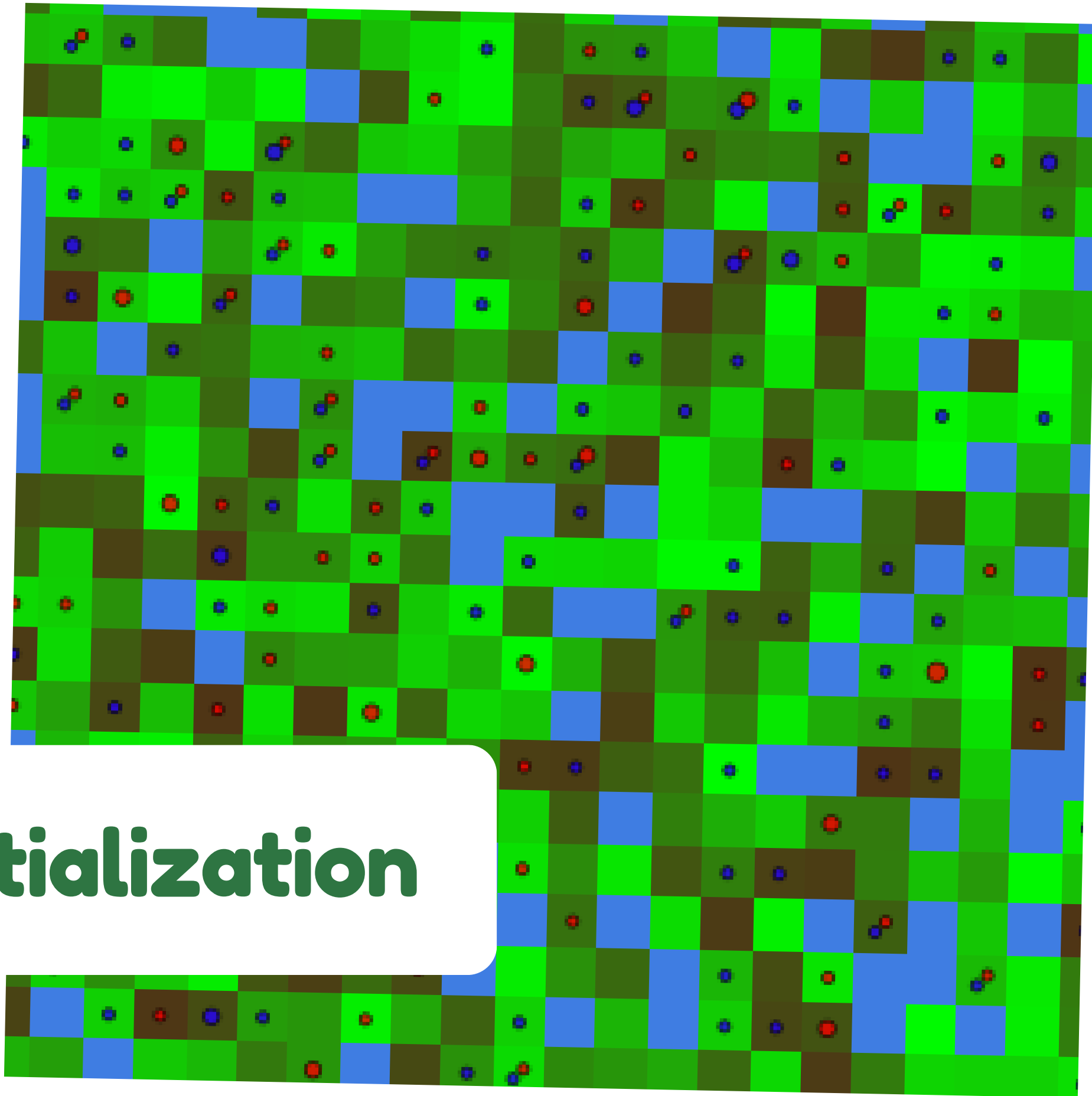
ERBAST:
Herbivores



CARVIZ:
Carnivores



A typical initialization



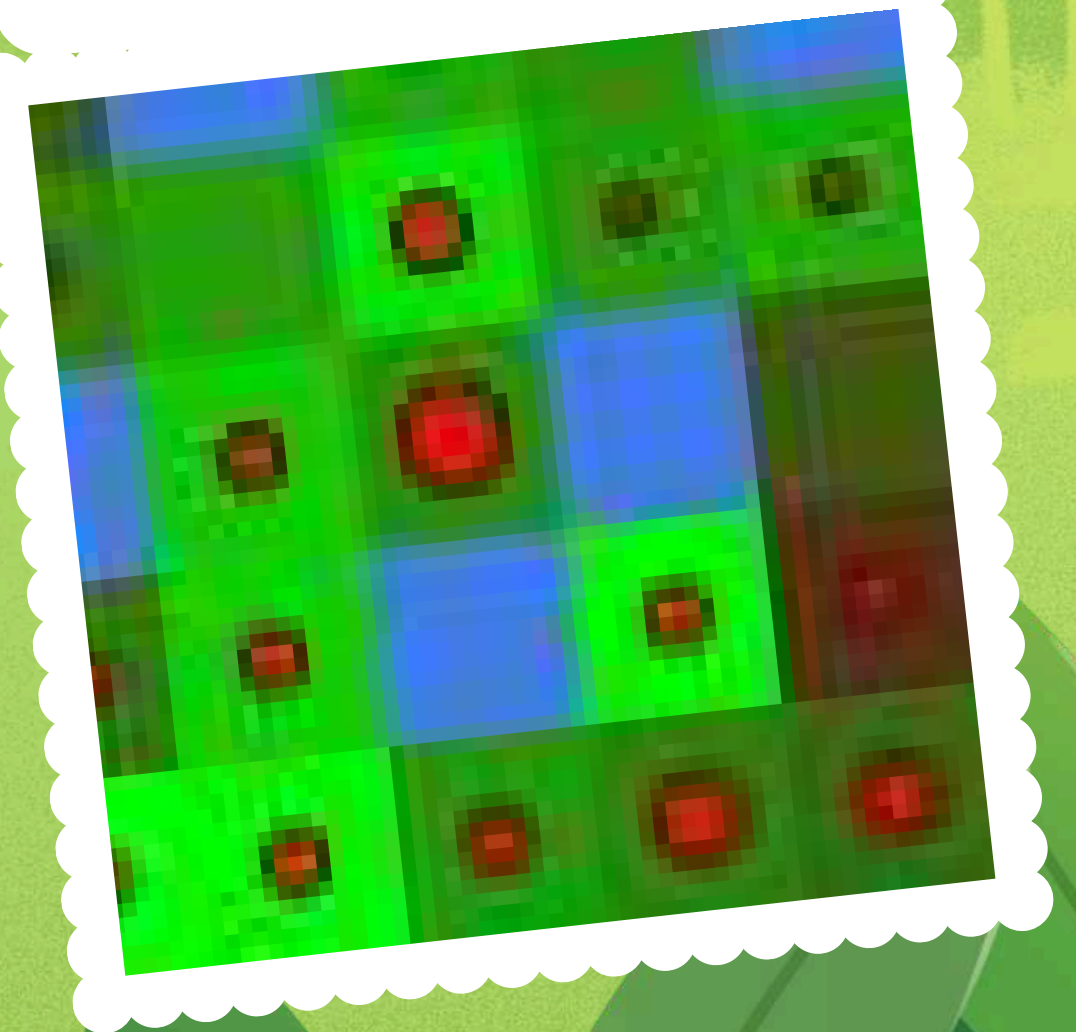
ALL BORDERS
ARE WATER



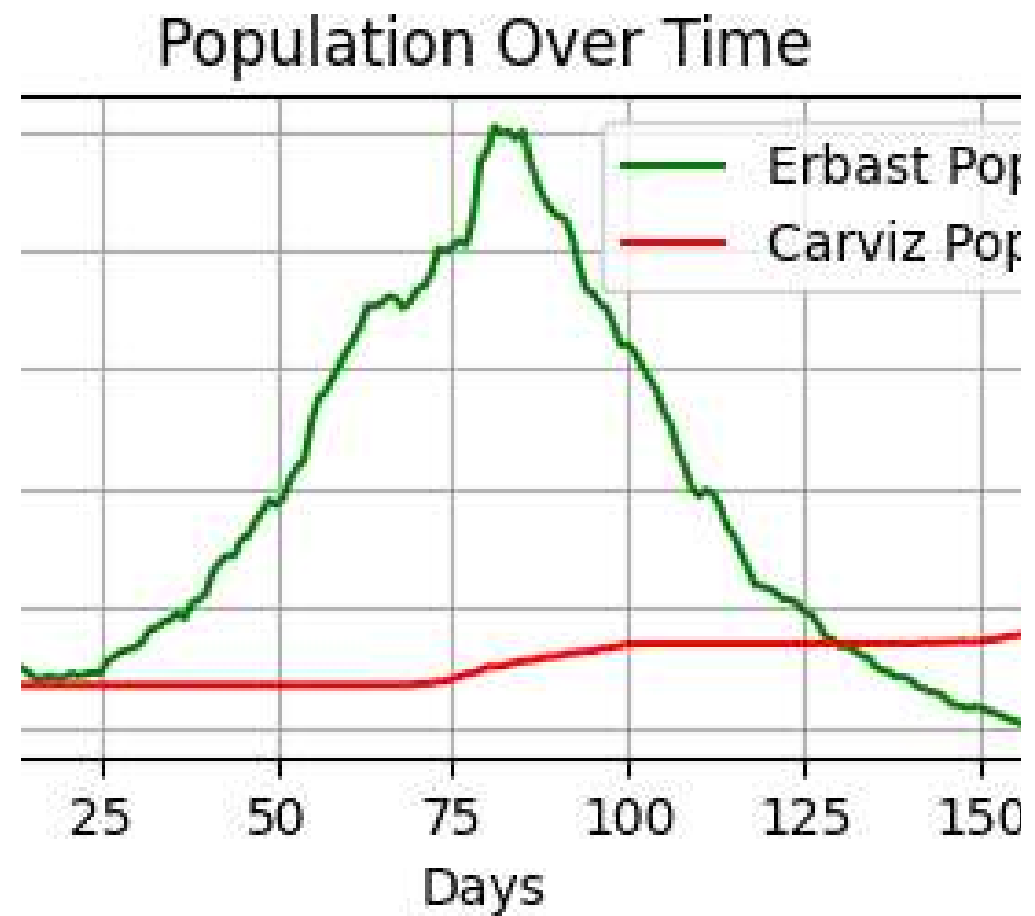
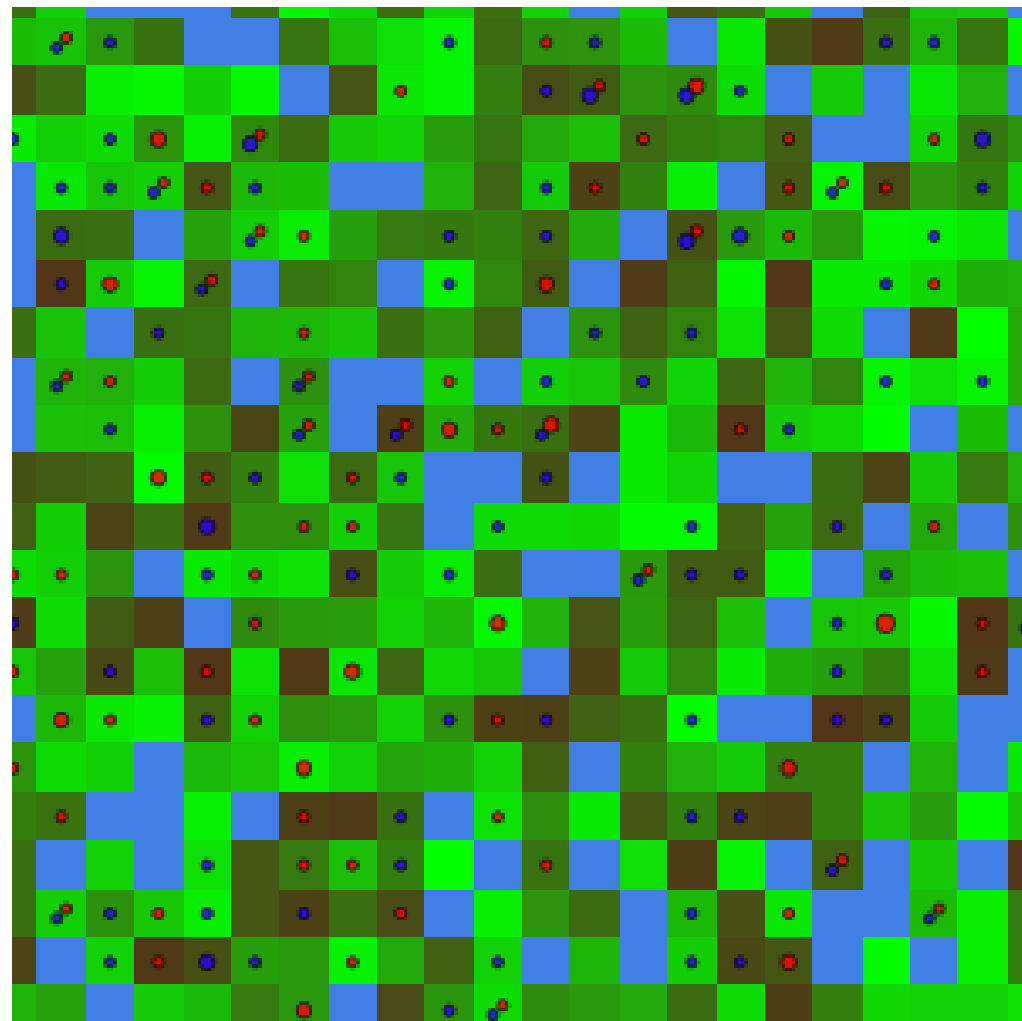
ERBAST FORM
HERDS



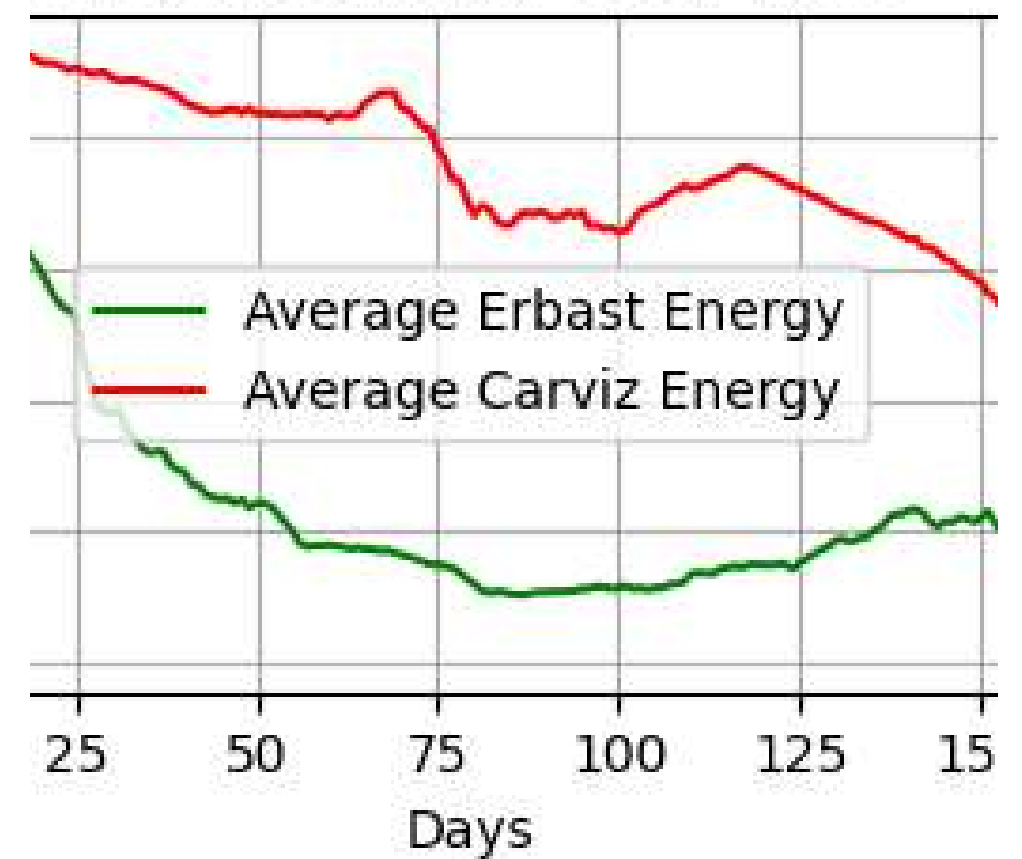
CARVIZ FORM
PRIDES



matplotlib



Average Energy Over Time



matplotlib



Initial Erbast



260

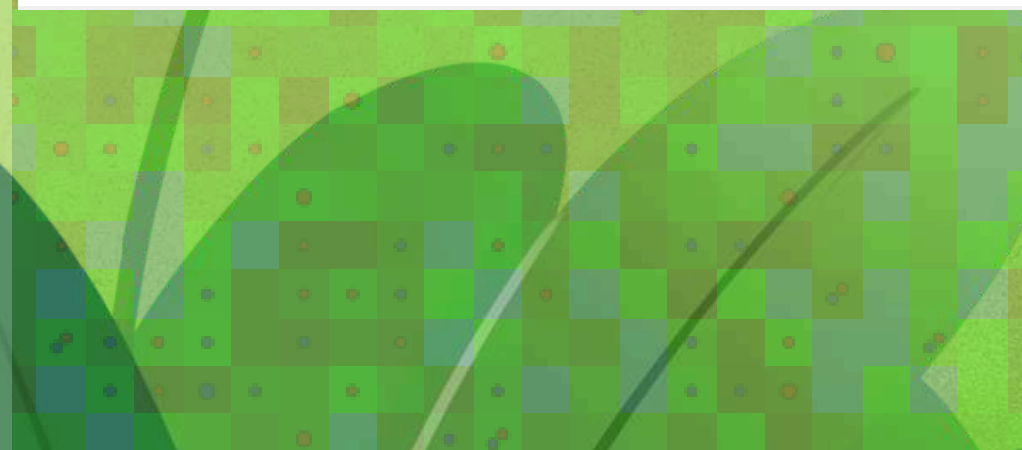
Initial Carviz



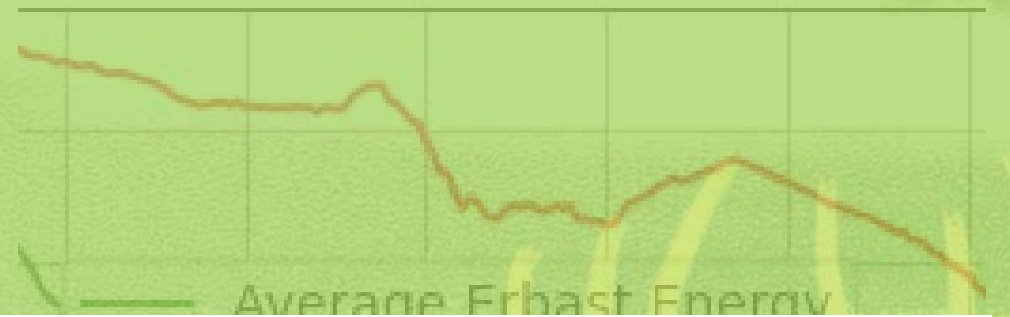
70

Pause/Resume

Reset

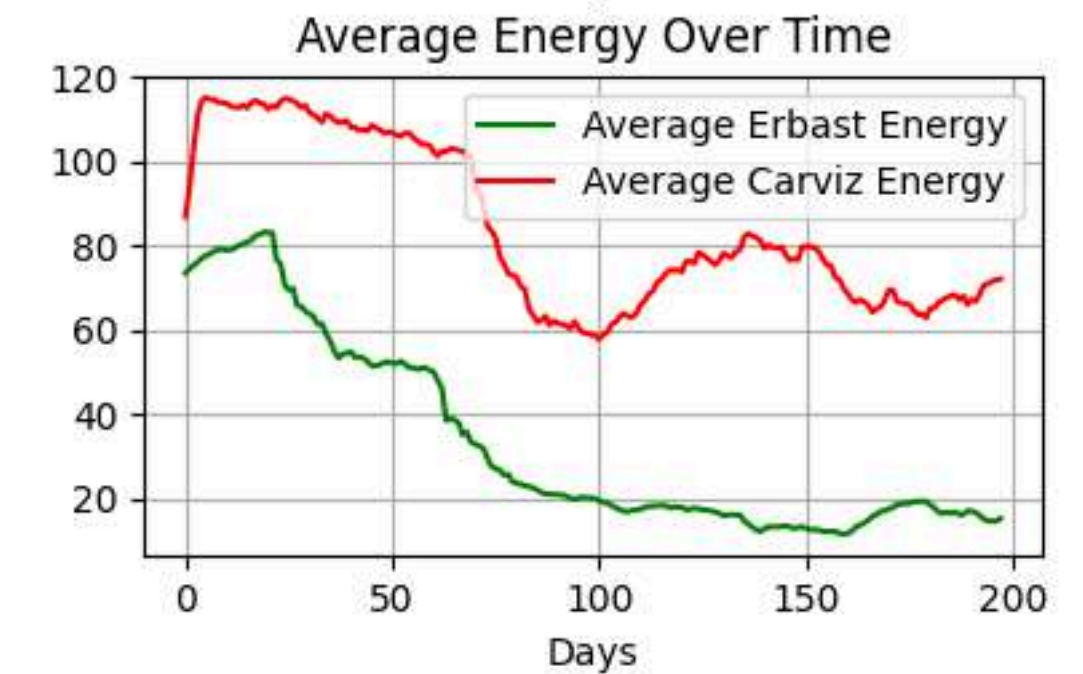
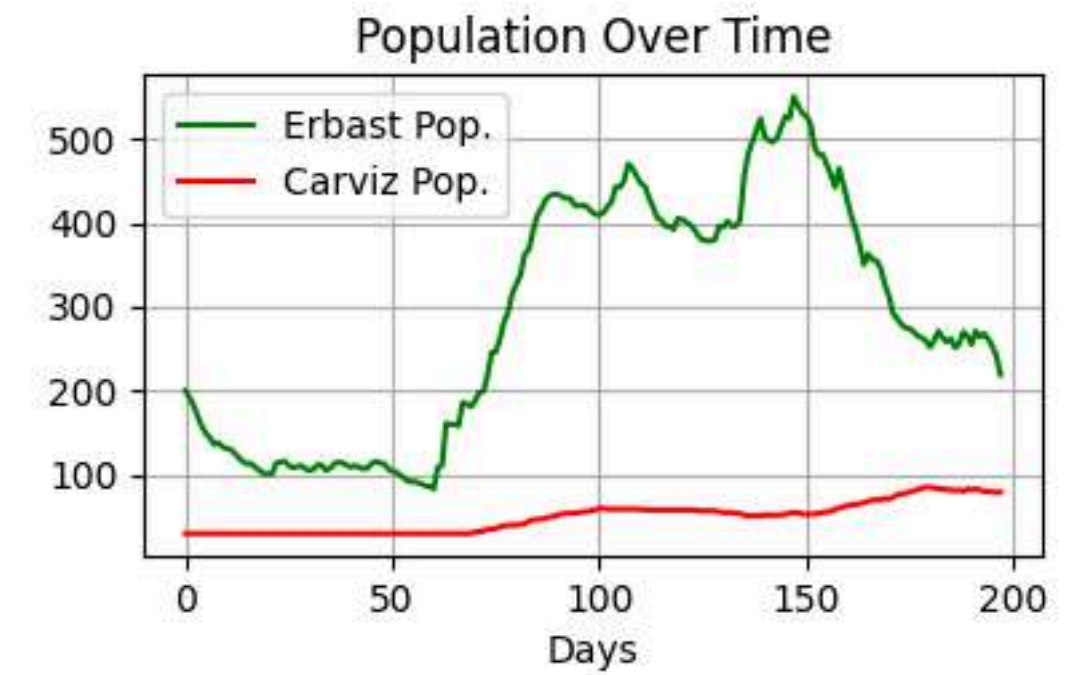
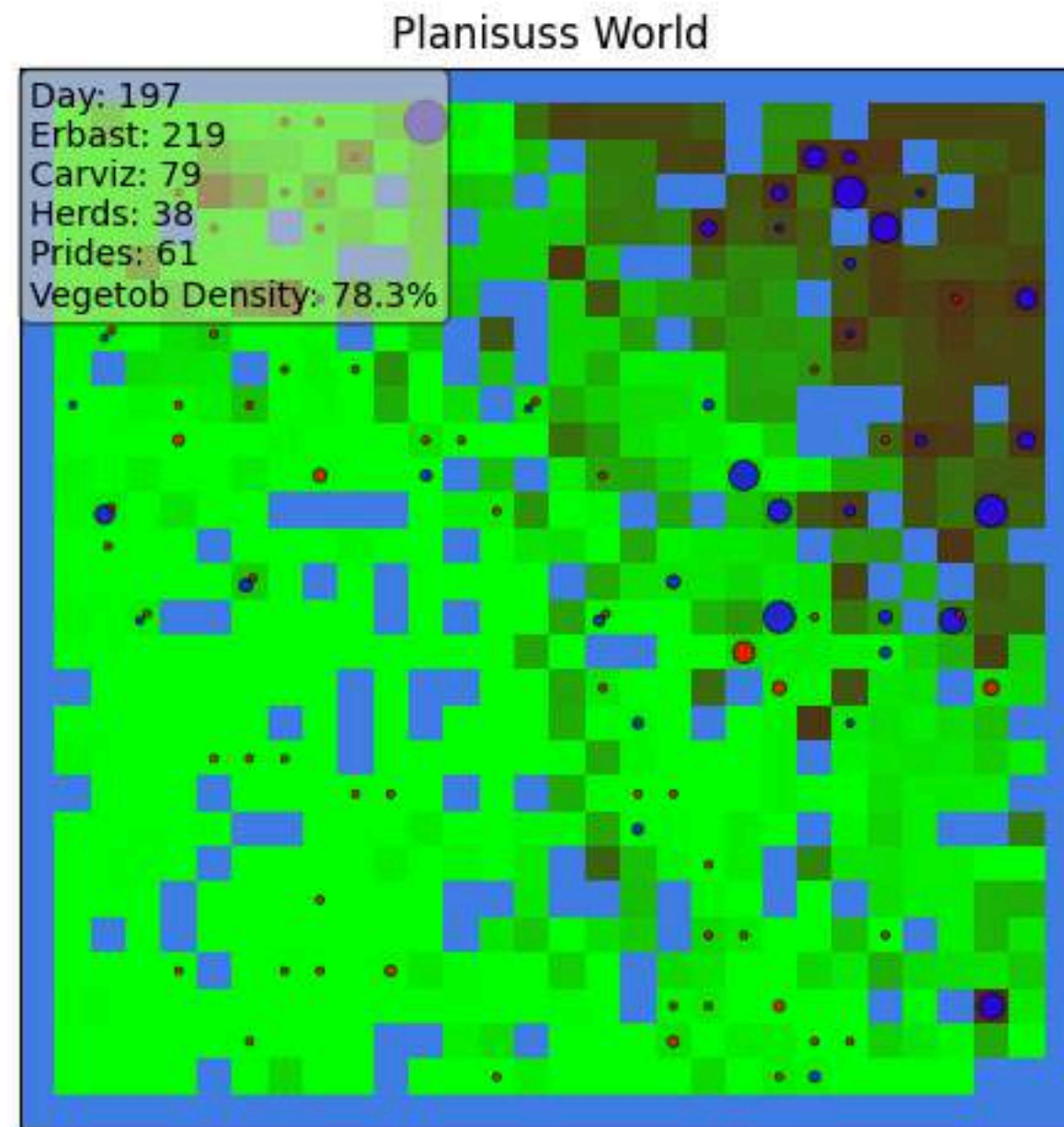


Average Energy Over Time



Day: 197
Erbast: 219
Carviz: 79
Herds: 38
Prides: 61
Vegetob Density: 78.3%

Planisuss World - Day: 197 (Paused)



Initial Erbast 200

Initial Carviz 30

Pause/Resume

Reset

A day in Planisuss



**GROWING
PHASE**

**ACTION
PHASE**

**STRUGGLE
PHASE**

**AGING &
SPAWNING
PHASE**

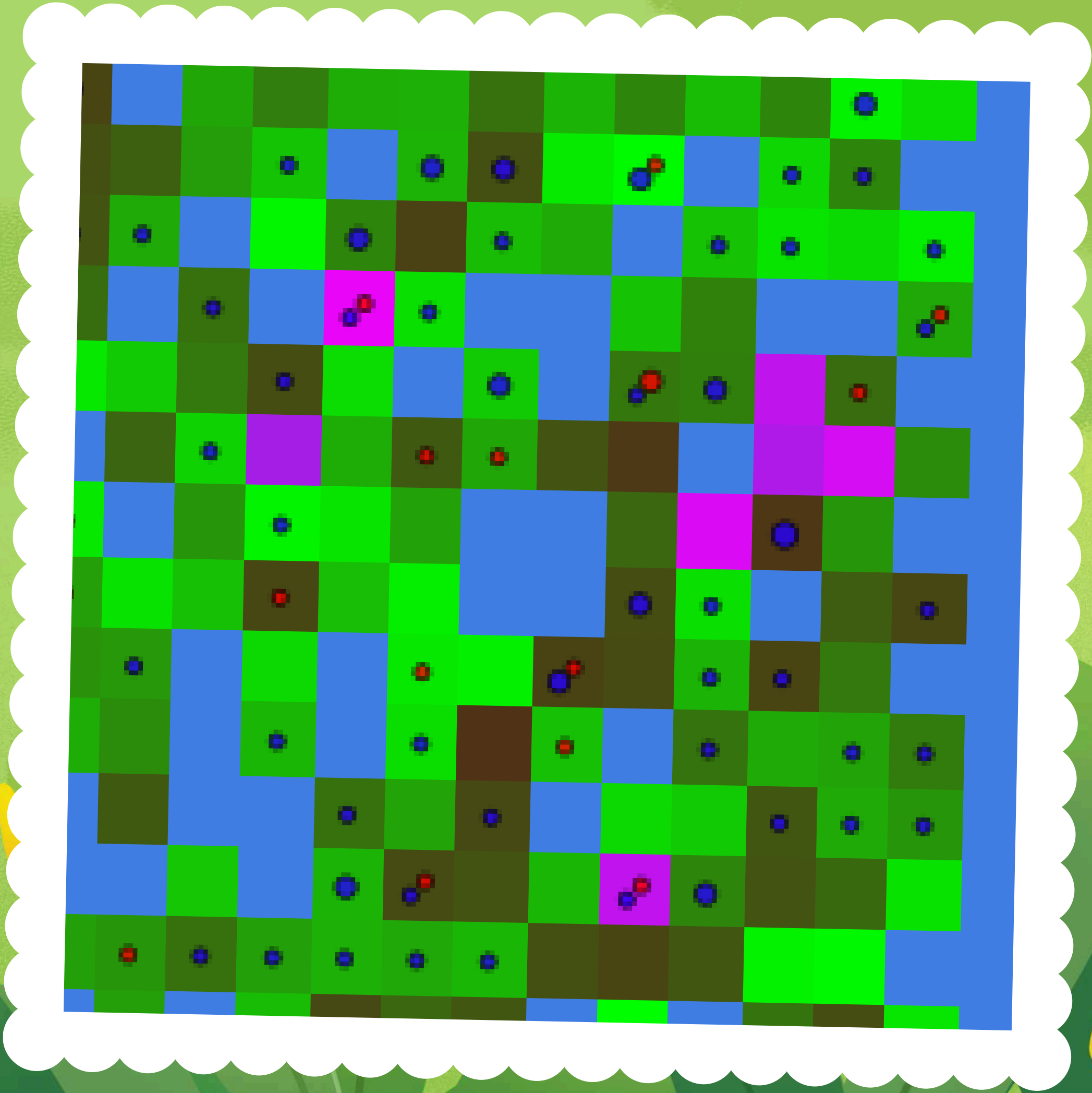
**GRAZING
PHASE**

CULLING PHASE

PERSONAL CONTRIBUTIONS

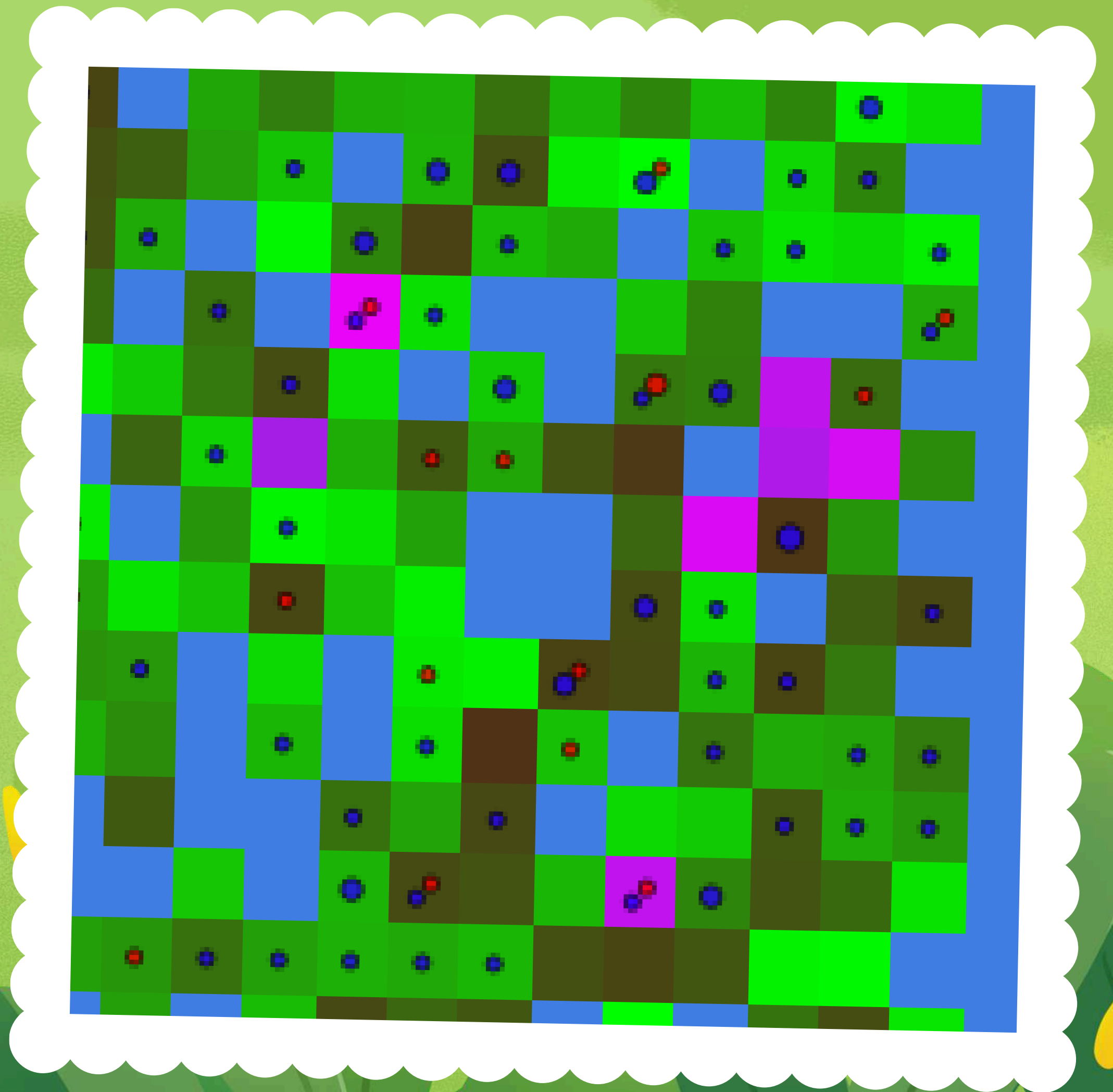


Terrain Types



A 15x15 grid map illustrating various terrain types. The map is enclosed in a white scalloped border. The terrain types are represented by different colors and symbols:

- Blue:** Represents water or a body of water, forming a large area on the right side of the map.
- Green:** Represents grass or forest, covering most of the central and left areas.
- Brown:** Represents dirt or a path, scattered throughout the map.
- Purple:** Represents a specific terrain type, appearing in a few scattered cells.
- Red and Blue Dots:** Small symbols scattered across the map, possibly representing resources or points of interest.





Escape Technique

**HERD
ATTACKS
PRIDE**

**A PORTION OF
THE HERD HAS A
CHANCE TO FLEE**

**HIGH ENERGY
CONSUMPTION**

**Randomness
introduced when
individual reaches
its lifetime value**

Spawning

```
if animal.age >= animal.lifetime:  
    child_energy1 = (animal.energy / 2) + random.randint(0, 10)  
    child_energy2 = (animal.energy / 2) + random.randint(0, 10)  
    newborns.append(Animal(animal.id, child_energy1))  
    newborns.append(Animal(animal.id, child_energy2))  
    continue # Parent dies after reproduction
```


Desperate search for food

**LOW
AVERAGE
ENERGY**

**SEARCH FOR
BETTER
CONDITIONS**

**ENERGY
CONSUMPTION
BUT HIGHER
GAIN LATER**



Desperate search for food



**SEARCH
FOR PREY
NEARBY**

**IF NO PREY IN
SIGHT, MOVE
RANDOMLY**

**ALLOWS FOR
EXPLORATION
OF BROADER
AREAS**

Prevents massive overpopulation by limiting the number of individuals in each cell

Culling Phase

```
def culling_phase(grid: Grid): # To handle overcrowding
    for i in range(grid.size):
        for j in range(grid.size):
            cell = grid[i, j]
            if len(cell.animals) > Config.CELL_CAPACITY:
                cell.animals.sort(key=lambda a: a.energy, reverse=True)
                cell.animals = cell.animals[:Config.CELL_CAPACITY]
```

Weakest are removed: NATURAL SELECTION



Thank you!

Federica Brasca